

Cue-induced cocaine craving: neuroanatomical specificity for drug users and drug stimuli.

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OBJECTIVE: Cocaine-related cues have been hypothesized to perpetuate drug abuse by inducing a craving response that prompts drug-seeking behavior. However, the mechanisms, underlying neuroanatomy, and specificity of this neuroanatomy are not yet fully understood. METHOD: To address these issues, experienced cocaine users (N=17) and comparison subjects (N=14) underwent functional magnetic resonance imaging while viewing three separate films that portrayed 1) individuals smoking crack cocaine, 2) outdoor nature scenes, and 3) explicit sexual content. Candidate craving sites were identified as those that showed significant activation in the cocaine users when viewing the cocaine film. These sites were then required to show significantly greater activation when contrasted with comparison subjects viewing the cocaine film (population specificity) and cocaine users viewing the nature film (content specificity). RESULTS: Brain regions that satisfied these criteria were largely left lateralized and included the frontal lobe (medial and middle frontal gyri, bilateral inferior frontal gyrus), parietal lobe (bilateral inferior parietal lobule), insula, and limbic lobe (anterior and posterior cingulate gyrus). Of the 13 regions identified as putative craving sites, just three (anterior cingulate, right inferior parietal lobule, and the caudate/lateral dorsal nucleus) showed significantly greater activation during the cocaine film than during the sex film in the cocaine users, which suggests that cocaine cues activated similar neuroanatomical substrates as naturally evocative stimuli in the cocaine users. Finally, contrary to the effects of the cocaine film, cocaine users showed a smaller response than the comparison subjects to the sex film. CONCLUSIONS: These data suggest that cocaine craving is not associated with a dedicated and unique neuroanatomical circuitry; instead, unique to the cocaine user is the ability of learned, drug-related cues to produce brain activation comparable to that seen with nondrug evocative stimuli in healthy comparison subjects.